

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester III Examination, 2023-24

Industrial Microbiology

Paper: IMB-301 — Agricultural and Food Microbiology

Time: Three Hours

Maximum Marks: 70

Instructions: Answer five questions in all, selecting two each from Sections A and B and Question 1 which is compulsory. The figures in the right-hand margin indicate marks.

Question 1. Answer the following in not more than 50 words each: [10 × 1 = 10]

- (a) What is associative mutualism?
- (b) Name an endotrophic bacterium that can fix nitrogen.
- (c) Write the name of a fungal species responsible for Turkey-X-Disease.
- (d) Define water activity.
- (e) Define modified atmosphere packaging.
- (f) Define blanching.
- (g) Name a mycotoxin which is nephrotoxic.
- (h) Write the full form of FSSAI.
- (i) Write the name of antimicrobial compounds present in *Allium sativum*.
- (j) Why is vermicompost better than compost?

Section-A

Question 2. Write an account of biofertilizers and their classification and discuss the advantages of liquid biofertilizers over carrier-based fertilizers. [15]

Question 3. Describe briefly the role of microbes in restoration of degraded lands. [15]

Question 4. Write short notes on any two of the following: [2 × 7.5 = 15]

- (a) Biofertilizer technology
- (b) Mutualistic nitrogen fixation
- (c) Worm casts

Question 5. Write short notes on any two of the following: [2 × 7.5 = 15]

- (a) Biological control of plant disease

- (b) Plant growth promoting micro-organisms
- (c) Mycorrhiza

Section-B

Question 6. Write an account of food-spoilage microbes and discuss their toxic effects on health. [15]

Question 7. Define the principle of food preservation and discuss the antimicrobial mechanism of action of chemical preservatives. [15]

Question 8. Write short notes on any two of the following: [2 × 7.5 = 15]

- (a) Single-cell protein
- (b) Medicinal importance of mushroom
- (c) Hurdle technology

Question 9. Write short notes on any two of the following: [2 × 7.5 = 15]

- (a) Intrinsic factors affecting growth and survival of food microbes
- (b) Fish spoilage microbes
- (c) Butter production

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